

R Worksheet: Matrices, Plots, Libraries, and Functions

Section 1: Matrices + Operations

Three students take a class together and study for a **project**, **midterm**, and **final** together.

- Student A scores: 12, 23, 34
- Student B scores: 45, 56, 67
- Student C scores: 78, 89, 64

Q1: Create a matrix containing their scores.

Use `matrix()` to create the 3x3 matrix:

```
# Solution:
m <- matrix(
  c(12, 45, 78,
    23, 56, 89,
    34, 67, 64),
  nrow = 3,
  ncol = 3,
  byrow = TRUE
)
```

Q2: Name the rows and columns

Give descriptive names to the columns (assignments) and rows (students) using `colnames()` and `rownames()`.

```
# Solution
colnames(m) <- c("Project", "Midterm", "Final")
rownames(m) <- c("Student A", "Student B", "Student C")
```

Q3: Compute each student's average score

Which student has the highest average?

```
# Solution
rowMeans(m)
```

Plots

Q4: How can we make simple graphs in Base R?

Solution:

Use `hist()`, `plot()`, and `boxplot()` to visualize data.

Q5: Create a histogram using the matrix scores

Hint: You can use `as.vector(m)` to convert matrix to a vector.

```
# Solution:
hist(as.vector(m))
```

Q6: Use the built-in `airquality` dataset

Let's use the `airquality` dataset, which has daily New York air quality measurements (May–Sept 1973). Variables include:

- `Ozone`: Ozone concentration (ppb)
- `Solar.R`: Solar radiation (langley)
- `Wind`: Wind speed (mph)
- `Temp`: Temperature (°F)
- `Month`: Month (5 = May, ..., 9 = September)
- `Day`: Day of the month

a) Histogram of Ozone (Hint: use `$` to access a column from a data frame)

```
# Solution:  
  
hist(airquality$Ozone)
```

b) Scatterplot: Ozone vs Temperature

```
# Solution:  
  
plot(airquality$Temp, airquality$Ozone,  
      xlab = "Temperature (°F)",  
      ylab = "Ozone (ppb)",  
      main = "Ozone vs Temperature")
```

c) Boxplot: Temperature by Month

```
# Solution:  
  
boxplot(Temp ~ Month, data = airquality,  
         xlab = "Month",  
         ylab = "Temperature (F)",  
         main = "Temperature by Month")
```

Intro to Libraries

Q7: Install and load the libraries

```
install.packages("igraph")  
install.packages("networkD3")  
  
library(igraph)  
library(networkD3)
```

Q8: What are `igraph` and `networkD3`?

How are they useful for data analysis and visualization?

Write lines of code to at the documentation:

```
# Solution:  
  
?igraph  
?networkD3
```

Q9: Explore usage

- `igraph` is used for creating and analyzing network graphs.
- `networkD3` helps build interactive network visualizations.

Run these examples on your own computer to understand each package's usage.

Example using `igraph`:

```
g <- graph(edges = c(1,2, 2,3, 3,1), n = 3, directed = FALSE)  
plot(g)
```

Example using `networkD3`:

```
simpleNetwork(data.frame(source = c("A", "B", "C"),  
                          target = c("B", "C", "A")))
```

Functions

Q10: Write a function to simulate rolling a die without replacement

```
# Solution:  
  
roll_die <- function(n = 6) {  
  sample(1:6, n, replace = FALSE)  
}  
  
roll_die()
```